Simulation code and parametrization of the figures

Bipolar and multipolar models of rules-based and power-based regimes

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1 Purpose

This note documents the code and the parameter values behind the figures listed in `Figuras_papers.pdf`: Figures 1.4, 1.5, 1.6, 1.9, 1.B.3, and 1.F.1 of the bipolar model, Figure 6 of the Papers and Proceedings (P&P) paper, which uses the two-leader variant of the bipolar model, and Figure 2.1 of the multipolar model, which is Figure 1 of the P&P paper. Table 1 maps each figure to the script that computes it, the data file it reads, and the image it produces. All files sit in the folder `figures_code`; the `README.md` there explains how to run them.

The bipolar figures are equilibrium maps: for every point of a grid of parameter values, a MATLAB script solves the model, records the symmetric Markov equilibrium, and saves the result to a `.mat` file. A separate plotting function draws the maps from the saved files, and two short Python scripts add labels and assemble panels. The multipolar figure plots closed-form expressions in Python and needs no solver.

Table 1: From figure to code

Figure	Solver script	Data file	Final image (`figures/`)
1.4 Symmetric Markovian equilibria	`main_RPC.m`	`main_2112.mat`	`fig_1_4_symmetric_markovian.png`
1.5 Efficient and inefficient equilibria	`main_RPC.m`	`main_2112.mat`	`fig_1_5_efficient_inefficient.png`
1.6 Rules-resistant and rules-compatible power-based equilibria	`main_RPC.m`	`main_2112.mat`	`fig_1_6_rules_resistant_compatible.png`
P&P Figure 6, Rules-based equilibria in a bipolar world (two-leader model)	`main_RPC_dotProduct_twoLeader.m`	`twoLeaders_1519.mat`	`fig_6_PP_bipolar_world.png`
1.9 Sustainability of a rules-lite regime, $c = 1.5$	`main_uRDelta.m`	`main_kappa15_c15.mat`	`fig_1_9_rules_lite_c15.png`
1.B.3 Enduring dominance	`main_RPC_changingQ22.m`	`prop4_{35,60,85,99}.mat`	`fig_1_B_3_enduring_dominance.png`
1.F.1 Equilibria under endogenous transitions, $\eta = \bar{\eta}$	`main_RPC_newEndogenous.m`	`endogenous_2213.mat`	`fig_1_F_1_endogenous_transitions.png`
2.1 Multipolar fragmentation and persistence (P&P Figure 1)	`persistence.py` (Python)	none	`fig_2_1_multipolar_persistence.pdf`

Notes: MATLAB scripts and data files are in `matlab/` and `matlab/data/`; `make_figures.m` draws every map from the data files. Figures 1.5 and 1.6 get their white labels from `legend_merger.py`; the four panels of Figure 1.B.3 are assembled by `image_merger.py`. The Python script for Figure 2.1 is in `python/`.

2 The bipolar model as coded

States, regimes, and strategies. Two countries, i and j, interact in discrete time and discount the future at rate δ. The state $s \in \{h_i, d_i, d_j, h_j\}$ records who leads: i is the hegemon at h_i, i is the dominant country of a bipolar world at d_i, and symmetrically for j. The code indexes the states 1 to 4 in this order. The regime $\rho \in \{P, R\}$ (power or rules) is a second state variable that persists until a country changes it. Under P at h_i, the hegemon builds rules with probability λ_1 and pays the building cost c once. Under R at d_i, the dominant country keeps the rules with probability λ_2; breaking them costs nothing and returns the regime to P. Country j uses the same (λ_1, λ_2) at h_j and d_j, so a symmetric Markov equilibrium is a pair (λ_1, λ_2). Two further restrictions are imposed as constraints: a hegemon never abandons rules that already exist, and no country builds rules in a dominance state.

Payoffs. Country i's per-period payoff depends on the state and the regime. Under rules it equals $u_R = 1$ in every state. Under power it equals

$$u^P(h_i) = 0, \qquad u^P(d_i) = 1 + k, \qquad u^P(d_j) = 1 - 2k, \qquad u^P(h_j) = 0,$$

with $k = 0.1$ in every figure. The joint per-period payoff is therefore 2 under rules, $2 - k = 1.9$ in a dominance state under power (the collective-action loss), and 0 in a hegemonic state under power. The building cost c is the horizontal axis of most maps.

Transitions. The state follows a Markov chain with transition matrix Q, rows and columns ordered (h_i, d_i, d_j, h_j):

$$Q = \begin{pmatrix} 0.10 & 0.90 & 0 & 0 \\ 0.45 & 0.10 & 0.45 & 0 \\ 0 & 0.45 & 0.10 & 0.45 \\ 0 & 0 & 0.90 & 0.10 \end{pmatrix}.$$

Hegemony is followed by the dominance of the same country with probability 0.9, and a dominant country either becomes hegemon or loses its lead with equal probability 0.45. The persistence of the dominance states, $q_{d_i,d_i} = q_{d_j,d_j} = 0.1$, is the parameter varied in Figure 1.B.3. Transitions do not depend on the regime except in Figure 1.F.1, where choosing power in a dominance state raises the chance of staying dominant.

Value functions. Let V_s^ρ denote i's value at state s under regime ρ at the start of a period, $V^\rho = (V_1^\rho, \ldots, V_4^\rho)'$, Q_s the s-th row of Q, and Q_s^P the corresponding row under power ($Q^P = Q$ except in Figure 1.F.1). The scripts solve the system

$$\begin{aligned}
V_1^P &= \lambda_1\big[u_1^R - c + \delta Q_1 V^R\big] + (1-\lambda_1)\big[u_1^P + \delta Q_1^P V^P\big],\\
V_1^R &= u_1^R + \delta Q_1 V^R,\\
V_2^P &= u_2^P + \delta Q_2^P V^P,\\
V_2^R &= \lambda_2\big[u_2^R + \delta Q_2 V^R\big] + (1-\lambda_2)\big[u_2^P + \delta Q_2^P V^P\big],\\
V_3^P &= u_3^P + \delta Q_3^P V^P,\\
V_3^R &= \lambda_2\big[u_3^R + \delta Q_3 V^R\big] + (1-\lambda_2)\big[u_3^P + \delta Q_3^P V^P\big],\\
V_4^P &= \lambda_1\big[u_4^R + \delta Q_4 V^R\big] + (1-\lambda_1)\big[u_4^P + \delta Q_4^P V^P\big],\\
V_4^R &= u_4^R + \delta Q_4 V^R,
\end{aligned}$$

where u_s^ρ is i's payoff at state s under ρ. At h_j it is j who pays the building cost, so i's value under power at h_j carries no cost term.

Equilibrium conditions. Write the one-shot deviation values at h_i under P (build or not) and at d_i under R (keep or break) as

$$D_1^P = u_1^P + \delta Q_1^P V^P, \qquad D_1^R = u_1^R - c + \delta Q_1 V^R,$$
$$D_2^P = u_2^P + \delta Q_2^P V^P, \qquad D_2^R = u_2^R + \delta Q_2 V^R.$$

A pair (λ_1, λ_2) together with (V^P, V^R) is an equilibrium when it solves the system above and satisfies four conditions:

(a) $\lambda_1 = 1$ if $D_1^R > D_1^P$, $\lambda_1 = 0$ if $D_1^R < D_1^P$, and $\lambda_1 \in (0,1)$ only if $D_1^R = D_1^P$;

(b) the same for λ_2 with D_2^R and D_2^P;

(c) $D_1^P \leq u_1^R + \delta Q_1 V^R$: at h_i under R the hegemon keeps the rules;

(d) $D_2^P \geq u_2^R - c + \delta Q_2 V^R$: at d_i under P the dominant country does not build rules.

Conditions (a) and (b) are the optimality of the two probabilities; (c) and (d) are the two restrictions on off-path behavior.

Classification of equilibria. The maps color each grid point by the equilibrium pair: $(\lambda_1, \lambda_2) = (1,1)$ is a rules-based equilibrium (the hegemon builds rules and the dominant country keeps them); $(1,0)$ is a cyclic equilibrium (rules are built under hegemony and broken as soon as the world turns bipolar); $(0,1)$ is a rules-compatible power-based equilibrium (nobody builds, but rules would be kept if they existed); $(0,0)$ is a rules-resistant power-based equilibrium (nobody builds and rules would be broken). Three-class maps merge the last two classes into "power-based equilibria".

Two-leader variant (P&P Figure 6). The script behind P&P Figure 6 adds a fifth state N between d_i and d_j: a multipolar state in which neither leader dominates and both act at once. The transition matrix, with rows and columns ordered (h_i, d_i, N, d_j, h_j), keeps the baseline rows for the hegemonic states and gives a dominant country equal chances 0.45 of becoming hegemon and of falling into N, where the world stays with probability 0.1 or moves to either dominance state with probability 0.45:

$$Q = \begin{pmatrix} 0.10 & 0.90 & 0 & 0 & 0 \\ 0.45 & 0.10 & 0.45 & 0 & 0 \\ 0 & 0.45 & 0.10 & 0.45 & 0 \\ 0 & 0 & 0.45 & 0.10 & 0.45 \\ 0 & 0 & 0 & 0.90 & 0.10 \end{pmatrix}.$$

At N the payoff depends on the pair of actions: 1 when both play rules, $1 - 2k = 0.8$ when i plays rules and j power, $1 + k = 1.1$ in the opposite case, and 0.95, the average of the two asymmetric outcomes, when both play power; under regime P both play power. A third probability λ_N gives the chance that a leader plays rules at N under regime R, with the joint-support assumption that rules survive there only if both leaders play rules, so the continuation under rules carries probability λ_N^2. The value equations and the four conditions above extend accordingly, with two additions: λ_N is optimal given the other leader's λ_N, and nobody builds rules at N under P. The script keeps, among multiple solutions, the one with $\lambda_2 = \lambda_N = 1$. In the three-class map, rules-based equilibria are $(\lambda_1, \lambda_2, \lambda_N) = (1,1,1)$, cyclic equilibria have $\lambda_1 = 1$ and rules broken at d_i or at N, and power-based equilibria have $\lambda_1 = 0$.

3 Numerical procedure and plotting

Grid. Each solver script evaluates the model on a grid of 201×201 equally spaced points, $\delta \in [0,1]$ (step 0.005) against $c \in [0,5]$ (step 0.025), or against $\gamma \in [0,0.1]$ (step 0.0005) in Figure 1.9. The last row and column ($\delta = 1$ and $c = 5$, where the system is singular or degenerate) are dropped before plotting, so every map shows a 200×200 grid with $\delta \leq 0.995$ and $c \leq 4.975$. The axis ticks are labeled with the nominal bounds.

Solver. At each grid point the script calls MATLAB's symbolic `solve` on the eight Bellman equations and the four equilibrium conditions of Section 2, with the ten unknowns $(V^P, V^R, \lambda_1, \lambda_2)$ and the restriction $\lambda_1, \lambda_2 \in [0,1]$. The loop over δ runs in parallel (`parfor`); a full map takes several hours. The solver returns every solution, including mixed strategies. The script stores (λ_1, λ_2) in two matrices, one row per δ and one column per c, and saves them to a `.mat` file.

Selection among multiple solutions. When the solver returns several equilibria at a grid point, the script keeps the one with $\lambda_2 = 1$ if it is unique. Cells with several solutions with $\lambda_2 = 1$, or none, are flagged with the codes -2 and -1, and cells where the solver fails keep `NaN`. In the seven data files behind the figures no such cell exists: every stored value lies in $[0,1]$.

Mixed strategies and the blurred boundary. In the final data λ_2 is always 0 or 1, while λ_1 takes interior values in a band along the boundary between the rules-based and the power-based regions. In that band no pure-strategy symmetric equilibrium exists: the dominant country would keep rules, and the hegemon is indifferent between building or not, so it builds with a probability that rises continuously with δ from 0 to 1. The band widens as c grows (for $c = 2$ it covers roughly $\delta \in [0.78, 0.88]$, for $c = 4$ it reaches the top of the map), which is why the green region fades into the red or orange region instead of ending at a sharp line. The one exception is the column $c = 1$, where the hegemon is exactly indifferent under cyclic play.

Colors. Each pixel's RGB color is an affine function of (λ_1, λ_2) with an interaction term, fitted so that the four pure profiles map to the four anchor colors in Table 2. For four-class maps,

$$
\begin{aligned}
R &= 0.8471 - 0.7294\,\lambda_1 + 0.0941\,\lambda_2 - 0.0549\,\lambda_1\lambda_2, \\
G &= 0.1059 + 0.2863\,\lambda_1 + 0.3647\,\lambda_2 - 0.1569\,\lambda_1\lambda_2, \\
B &= 0.2510 + 0.5333\,\lambda_1 - 0.0157\,\lambda_2 - 0.3686\,\lambda_1\lambda_2,
\end{aligned}
$$

and for three-class maps the coefficients on λ_2 are replaced so that $(0,1)$ and $(0,0)$ both map to red ($R = 0.8471 - 0.7294\,\lambda_1 + 0.0392\,\lambda_1\lambda_2$, $G = 0.1059 + 0.2863\,\lambda_1 + 0.2078\,\lambda_1\lambda_2$, $B = 0.2510 + 0.5333\,\lambda_1 - 0.3843\,\lambda_1\lambda_2$). Mixed values of λ_1 therefore appear as blends of the anchor colors.

Table 2: Anchor colors of the equilibrium maps

Equilibrium class	(λ_1, λ_2)	Color	RGB
Rules-based	$(1,1)$	green	$(38, 153, 102)$
Cyclic	$(1,0)$	blue	$(30, 100, 200)$
Rules-compatible power-based	$(0,1)$	orange (four-class maps only)	$(240, 120, 60)$
Rules-resistant power-based	$(0,0)$	red	$(216, 27, 64)$

Notes: In three-class maps the rules-compatible class is drawn in red as well. The same green and blue are used for the regions of Figure 2.1.

Output and post-processing. `make_figures.m` loads each data file and calls `plot_map.m` (or `plot_map_gamma.m` for Figure 1.9), which builds the RGB image, draws it with `imshow`, adds the axes and the legend, and writes a PNG of 1094×973 pixels. The white labels of Figures 1.5 and 1.6 are written on those PNGs by `legend_merger.py`, and the four panels of Figure 1.B.3 are cropped, titled, and stacked by `image_merger.py`, which renders the panel titles with LaTeX.

Independent check. The script `python/check_pure_equilibria.py` recomputes the pure-strategy classification of every map with plain linear algebra: for each grid point and each pure pair it solves the Bellman system, checks conditions (a) to (d), and applies the selection rule. It agrees with the saved data on every cell where the MATLAB solver found a pure strategy, for all seven data files of the four-state model, which confirms the reading of the model given above and the parameter values reported in Section 4. The companion script `python/check_two_leaders.py` does the same for the two-leader model of P&P Figure 6 and agrees on all pure cells but two, which the MATLAB solver itself flagged as ambiguous and which are drawn red either way.

4 The figures

Table 3 collects the parameter values of every figure. Unless a figure states otherwise, it uses the baseline of Section 2: $k = 0.1$, $u_R = 1$, the transition matrix Q above, and the grid of Section 3.

Table 3: Parameter values by figure

Figure	Axes	Departure from the baseline
1.4	$c \in [0,5]$, $\delta \in [0,1]$	none
P&P 6	$c \in [0,5]$, $\delta \in [0,1]$	two-leader variant: fifth state N, 5×5 transition matrix, payoffs at N by action pair, third probability λ_N (Section 2)
1.5	same data as 1.4	white curve $c = f(\delta)$, the planner's threshold (Section 4.2)
1.6	same data as 1.4	four-class colors and the same white curve
1.9	$\gamma \in [0, 0.1]$, $\delta \in [0,1]$	$c = 1.5$; rules payoffs $(1+\gamma,\ 1+\gamma,\ 1-\kappa\gamma,\ 1-\kappa\gamma)$ at (h_i, d_i, d_j, h_j) with $\kappa = 1.5$
1.B.3	$c \in [0,5]$, $\delta \in [0,1]$	$q_{d_i,d_i} = q_{d_j,d_j} = q \in \{0.35, 0.60, 0.85, 0.99\}$, with rows $Q_{d_i} = (\frac{1-q}{2}, q, \frac{1-q}{2}, 0)$ and $Q_{d_j} = (0, \frac{1-q}{2}, q, \frac{1-q}{2})$
1.F.1	$c \in [0,5]$, $\delta \in [0,1]$	under power, $Q^P_{d_i} = (0.45, 0.10+\eta, 0.45-\eta, 0)$ and $Q^P_{d_j} = (0, 0.45-\eta, 0.10+\eta, 0.45)$ with $\eta = \bar{\eta} = 0.45$; under rules, Q as in the baseline
2.1 (P&P 1)	$q_{0,0} \in [0,1]$	multipolar model: $c = 8$, $\delta = 0.4$, $N \in \{3, 6\}$, $\varphi(N) = 1.52\, e^{-0.165N}$

4.1 Figure 1.4, Symmetric Markovian equilibria

`main_RPC.m` solves the baseline model on the (c, δ) grid and saves `main_2112.mat`; `make_figures.m` draws it with three classes and no overlay (Figure 1).

Three features of the map follow directly from the model. First, the cyclic region (blue) lies to the left of $c = 1$. Under cyclic play the rules built at h_i are broken as soon as the world turns bipolar, so the hegemon's return to building is essentially the static gain $u_R - u^P(h_i) - c = 1 - c$, which is positive only if $c < 1$; the exact boundary is $c\,(1 - q_{h_i,h_i}\delta) = 1$, within one grid step of the vertical line. Second, for $c > 1$ building pays only if the rules survive the bipolar phase, which

requires the dominant country to keep them ($\lambda_2 = 1$) and the hegemon to be patient enough to recoup c; the rules-based region (green) is therefore bounded below by a δ threshold that rises with c, and the transition is the blurred band of mixed strategies described in Section 3. Third, the boundary between the cyclic and the rules-based regions for $c < 1$ rises as c falls: when rules are cheap to rebuild, a dominant country loses little by breaking them, so more patience is needed to keep them. Everything below these thresholds is power-based (red).

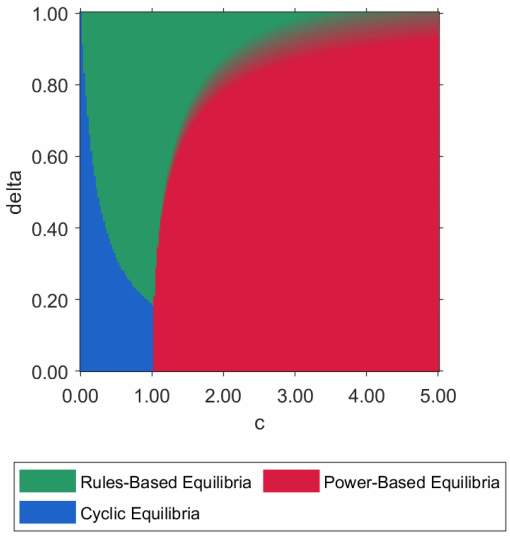


Figure 1: Figure 1.4, Symmetric Markovian equilibria (baseline parameters)

Notes: `main_RPC.m` → `main_2112.mat` → `plot_map(..., 'plot_RPC.png', false, false)`.

4.2 Figure 1.5, Efficient and inefficient equilibria

The figure overlays on the data of Figure 1.4 the locus of building costs at which a social planner is indifferent between permanent rules and permanent power (Figure 2). Let W_h and W_d denote the joint discounted payoff of the two countries under permanent power, starting from a hegemonic and a dominance state. With joint per-period payoffs of 0 in hegemonic states and 1.9 in dominance states, the baseline transitions give

$$W_h = \delta\,(0.1\,W_h + 0.9\,W_d), \qquad W_d = 1.9 + \delta\,(0.45\,W_h + 0.55\,W_d),$$

so that $W_h = 1.71\,\delta/(1 - 0.65\,\delta - 0.35\,\delta^2)$. Permanent rules built in a hegemonic state are worth $2/(1-\delta) - c$, so building is efficient if and only if $c \le f(\delta)$, where

$$f(\delta) = \frac{2}{1-\delta} - \frac{1.71\,\delta}{1 - 0.65\,\delta - 0.35\,\delta^2}.$$

The white curve is $c = f(\delta)$, coded in `plot_map.m` and drawn when the argument `social_planner` is true; it starts at $c = 2$ for $\delta = 0$ and grows without bound as $\delta \to 1$. Power-based equilibria to the left of the curve are inefficient, because a planner would build rules. Cyclic equilibria are inefficient everywhere, because keeping rules costs nothing and raises the joint payoff by k in every bipolar period. The two white labels are written on the PNG by `legend_merger.py`.

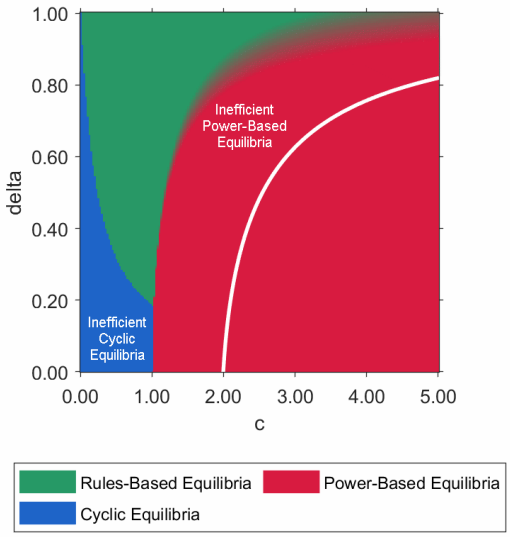


Figure 2: Figure 1.5, Efficient and inefficient equilibria

Notes: `main_2112.mat` → `plot_map(..., 'plot_socialPlanner.png', true, false)` → `legend_merger.py`.

4.3 Figure 1.6, Rules-resistant and rules-compatible power-based equilibria

Same data and same white curve as Figure 1.5, drawn with four classes (Figure 3): the power-based region is split by λ_2 into rules-compatible equilibria (orange, the dominant country would keep rules if they existed) and rules-resistant equilibria (red, it would break them). Because no rules are built along the path of a power-based equilibrium, c does not enter the dominant country's comparison, and the boundary between the two classes is horizontal at the δ at which the static gain from breaking rules, $k = 0.1$, equals the discounted loss from a bipolar world without rules (about $\delta = 0.17$ in the baseline). The labels "Inefficient RCP Equilibria" and "Inefficient RRP Equilibria" are written by `legend_merger.py`.

4.4 Figure 1.9, Sustainability of a rules-lite regime with $c = 1.5$

`main_uRDelta.m` fixes $c = 1.5$ and replaces the building cost on the horizontal axis by a tilt $\gamma \in [0, 0.1]$ of the rules in favor of the leading country (the code calls it `l`). Under rules the hegemon and the dominant country receive $1 + \gamma$, and the follower receives $1 - \kappa\gamma$ at d_j and h_j, with $\kappa = 1.5$; payoffs under power are the baseline ones. With $\kappa > 1$ the tilt is not a pure transfer: it lowers the joint payoff under rules by $(\kappa - 1)\gamma$ per period. The script saves `main_kappa15_c15.mat` (the name encodes 10κ and $10c$), and `plot_map_gamma.m` draws it with four classes (Figure 4). The legend lists three classes because no cyclic equilibrium exists at $c = 1.5 > 1$.

The rules-resistant region at the bottom shrinks as γ grows and vanishes at $\gamma = k = 0.1$, where the dominant country's static gain from breaking the rules, $k - \gamma$, reaches zero. The lower boundary of the rules-based region declines with γ because the hegemon's gain from building rules, $1 + \gamma - c$ per period of hegemony, rises with the tilt. The other runs of the same script kept in the working folder ($\kappa \in \{1, 2\}$, $c \in \{0.1, 0.5\}$) were explorations and are not in the paper.

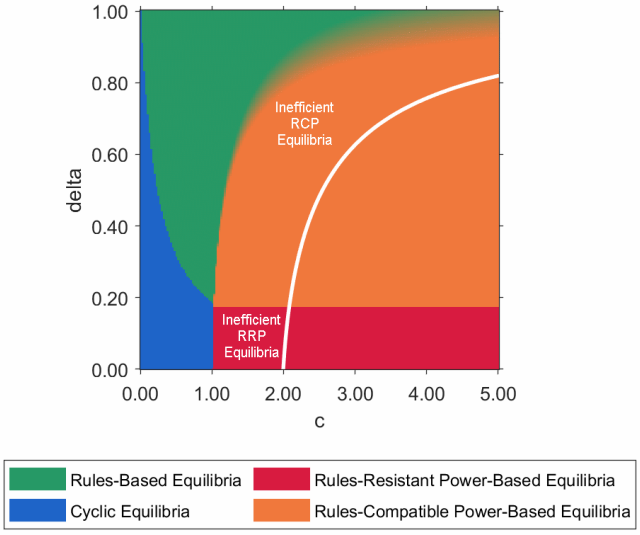


Figure 3: Figure 1.6, Rules-resistant and rules-compatible power-based equilibria

Notes: `main_2112.mat` → `plot_map(..., 'plot_socialPlanner_fourClasses.png', true, true)` → `legend_merger.py`.

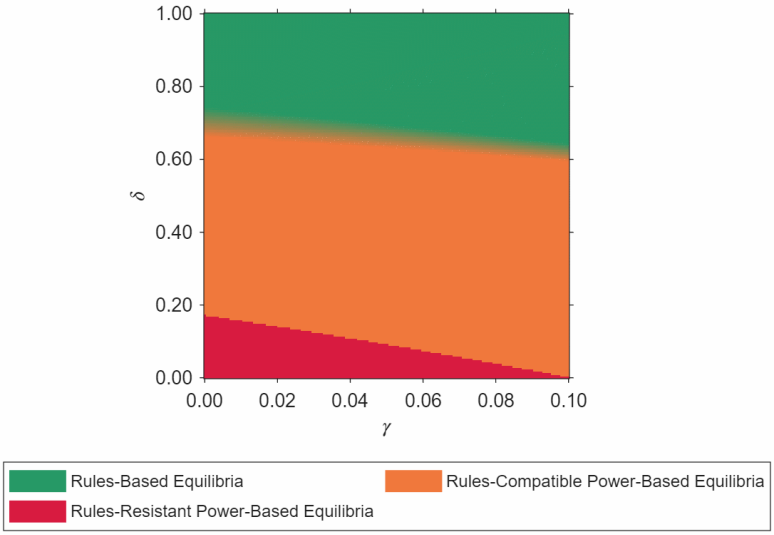


Figure 4: Figure 1.9, Sustainability of a rules-lite regime with $c = 1.5$

Notes: `main_uRDelta.m` ($c = 1.5$, $\kappa = 1.5$) → `main_kappa15_c15.mat` → `plot_map_gamma(..., 'rules_lite_c15.png', false, true)`.

4.5 Figure 1.B.3, Enduring dominance

`main_RPC_changingQ22.m` runs the baseline solver four times, once for each persistence of the dominance states $q \in \{0.35, 0.60, 0.85, 0.99\}$. It sets $q_{d_i,d_i} = q_{d_j,d_j} = q$ and splits the remaining probability equally between the two neighboring states, so the rows of d_i and d_j become $(\frac{1-q}{2}, q, \frac{1-q}{2}, 0)$ and $(0, \frac{1-q}{2}, q, \frac{1-q}{2})$; the hegemonic rows and all payoffs are the baseline ones. Each run saves `prop4_XX.mat` with $XX = 100q$ (the script is named after Proposition 4), `make_figures.m` draws the four three-class maps, and `image_merger.py` crops them, adds the LaTeX titles $q_{d_i,d_i} = \ldots$, and stacks them with a single legend (Figure 5).

As dominance becomes more persistent the rules-based region recedes toward $\delta = 1$ and the cyclic region grows: a dominant country that expects to stay dominant for long has little to gain from the rules it would enjoy as a follower, so it breaks them unless it is very patient. At $q = 0.99$ no rules-based equilibrium remains on the grid; the map shows cyclic equilibria for $c < 1$ and power-based equilibria otherwise.

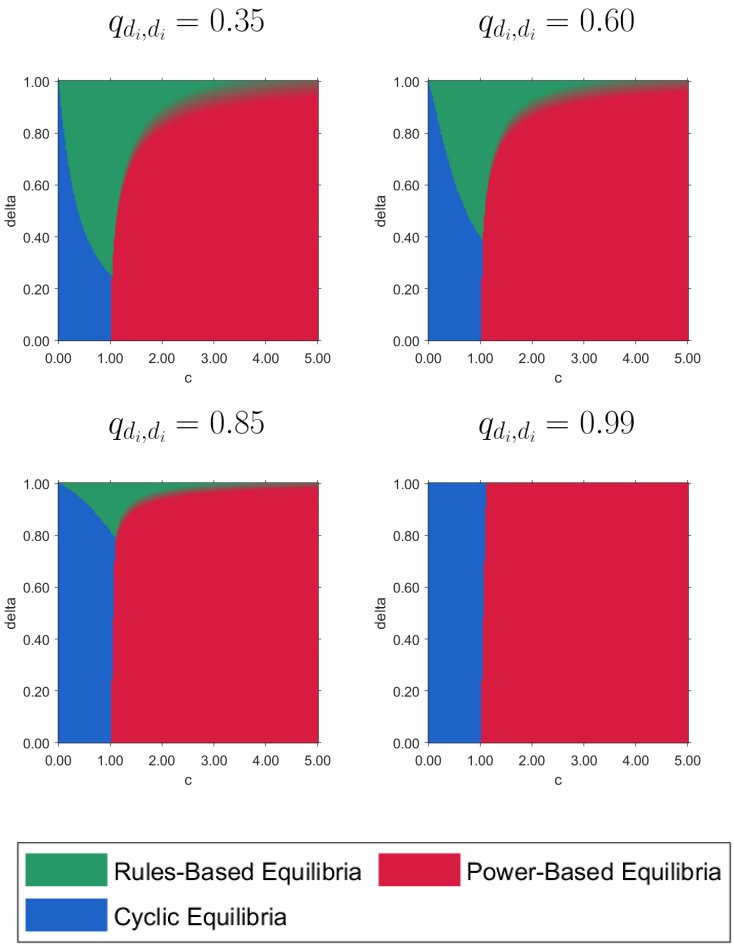


Figure 5: Figure 1.B.3, Enduring dominance

Notes: `main_RPC_changingQ22.m` → `prop4_{35,60,85,99}.mat` → `plot_map(..., 'plot_prop4_XX.png', false, false)` → `image_merger.py`.

4.6 Figure 1.F.1, Equilibria under endogenous transitions, $\eta = \bar{\eta}$

`main_RPC_newEndogenous.m` lets the choice of power in a dominance state raise the chance of remaining dominant. The script's variable `gamma` is the paper's η: under power, probability mass η moves from q_{d_i,d_j} to q_{d_i,d_i} (and symmetrically at d_j), so the rows of d_i and d_j become $(0.45, 0.10+\eta, 0.45-\eta, 0)$ and $(0, 0.45-\eta, 0.10+\eta, 0.45)$, while under rules the baseline rows apply. The figure uses $\eta = \bar{\eta} = 0.45$, the largest shift that keeps every probability nonnegative, so a dominant country that plays power never hands its lead to the other country directly. The modified rows enter every continuation value under power at d_i and d_j, including the deviation values in conditions (b) and (d). The script saves `endogenous_2213.mat`, and `make_figures.m` draws it with four classes and no overlay (Figure 6).

Compared with Figure 1.6, the rules-resistant region is larger (its upper boundary moves from about $\delta = 0.17$ to about 0.22), because power now also buys persistence of the dominant position. All boundaries are sharp: the solver found a pure-strategy equilibrium at every grid point of this run.

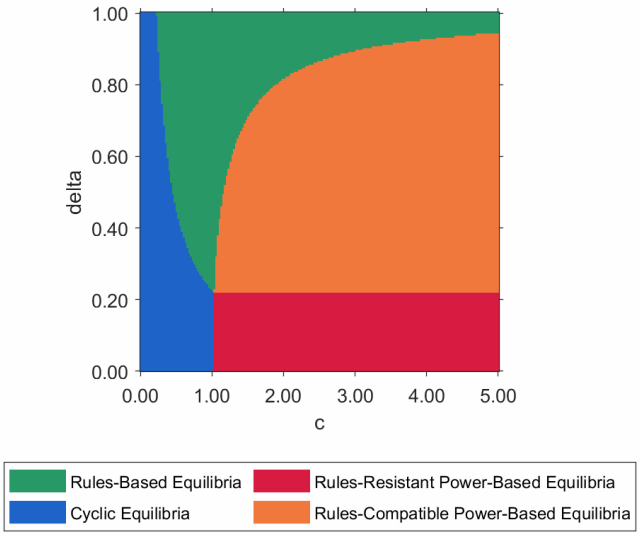


Figure 6: Figure 1.F.1, Equilibria under endogenous transitions, $\eta = \bar{\eta}$

Notes: `main_RPC_newEndogenous.m` ($\eta = 0.45$) → `endogenous_2213.mat` → `plot_map(..., 'plot_endogenous.png', false, true)`.

4.7 P&P Figure 6, Rules-based equilibria in a bipolar world

`main_RPC_dotProduct_twoLeader.m` solves the two-leader variant of Section 2 on the same (c, δ) grid, with $k = 0.1$, $u_R = 1$, and the five-state transition matrix, and saves `twoLeaders_1519.mat`, which holds the three strategy matrices. `make_figures.m` draws it with `plot_map_twoLeaders.m` as a three-class map and no overlay (Figure 7). The figure looks like Figure 1.4 but is a different run: rules now also have to survive the multipolar state, where each leader gains k in the short run from playing power and the regime collapses unless both refrain. The cyclic region is therefore larger

(it reaches about $\delta = 0.45$ at $c = 1$, against 0.2 in Figure 1.4) and the rules-based region starts at higher δ for every c. The solver found mixed strategies in a thinner band than in the baseline.

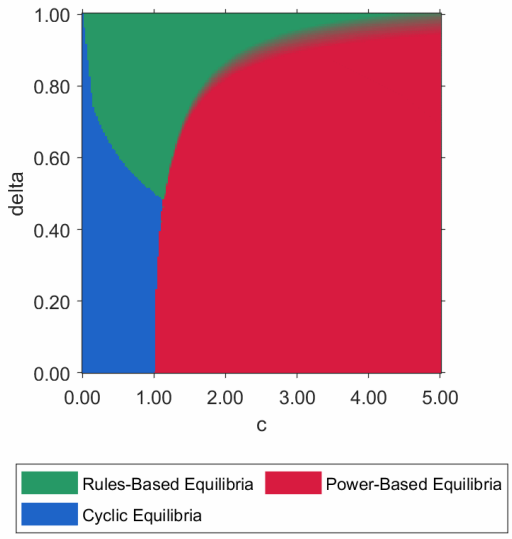


Figure 7: P&P Figure 6, Rules-based equilibria in a bipolar world (two-leader model)

Notes: `main_RPC_dotProduct_twoLeader.m` → `twoLeaders_1519.mat` → `plot_map_twoLeaders(..., 'plot_twoLeaders_three.png', false, false)`.

4.8 Figure 2.1, Multipolar fragmentation and persistence (P&P Figure 1)

This figure belongs to the multipolar model of the P&P paper. The Python script `persistence.py` produces it without any solver. With symmetric transitions out of the multipolar state, so that $q_{0,i} = (1 - q_{0,0})/N$ for every country i, the paper's sustainability condition for rules compares a dynamic benefit with a static cost. The script plots, as functions of the multipolar persistence $x = q_{0,0} \in [0, 1]$,

$$\begin{aligned} \text{efficiency advantage} &= \frac{\delta x}{1 - \delta x}, \\ \text{hegemony prospects} &= \frac{c\,\delta}{N}\,\frac{1 - x}{1 - \delta x}, \\ \text{total benefit} &= \text{efficiency advantage} + \text{hegemony prospects}, \\ \text{total cost} &= \varphi(N) = 1.52\, e^{-0.165N}, \end{aligned}$$

with $c = 8$, $\delta = 0.4$, and one panel for each $N \in \{3, 6\}$. The cost $\varphi(N)$ is the one-shot deviation gain under the payoff specification of the paper's footnote, where multipolar payoffs are $2n_i/(N-1) - 1$ under rules and $(n_i/(N-1))(1 + \varphi(N))$ under power, n_i being the number of other countries choosing rules. The bar along the horizontal axis is green where the total benefit exceeds the cost (a rules-based equilibrium coexists with the cyclic one) and blue where it does not (only the cyclic equilibrium exists). The script writes `persistence.pdf`, which the P&P manuscript includes as

`Figure1.pdf` (Figure 8); the colors are the green and blue of the maps and the font is Times New Roman.

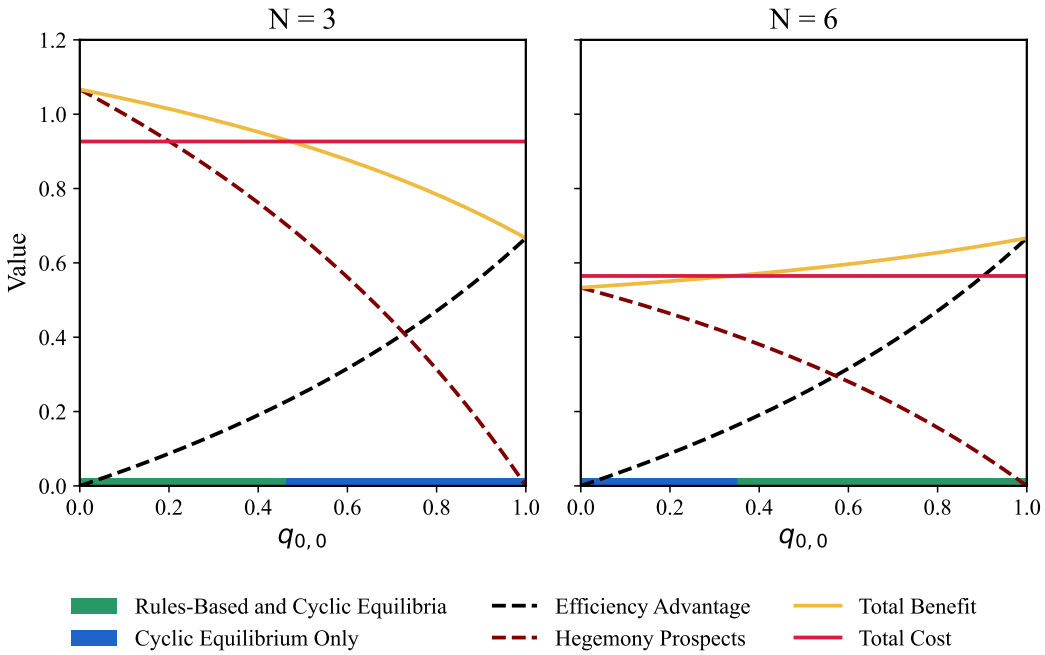


Figure 8: Figure 2.1, Multipolar fragmentation and persistence effects on rules-based equilibria

Notes: `python/persistence.py` ($c = 8$, $\delta = 0.4$, $N \in \{3, 6\}$) $\rightarrow$ `persistence.pdf`.